L7: Knowledge Check

1 Multiple choice 1 point

Which of the following statements about graph partitioning and community detection is/are **TRUE**:

- In graph partitioning the goal is to split the network in equal-sized subgraphs.
- In community detection the number of communities is given.
- Graph partitioning and community detection are two names given to the same problem by different people.
- Community detection was a precursor to Graph partitioning and influenced its development

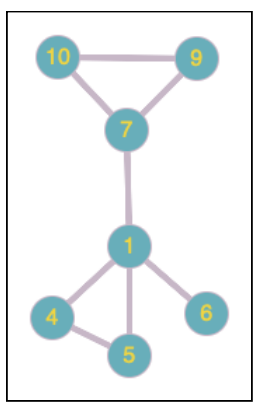
2 Multiple choice 1 point

Which of the following statements about network communities and community detection is/are **TRUE** :

- A community often spans two isolated components
- The distance between two nodes in different communities should be larger than the distance between two nodes in the same community
- The edge density within a community is greater than the edge density between communities.
- Every network can be partitioned in two or more communities.

3 Multiple choice 1 point

Compute the similarity between the nodes 1 and 5 in the given network.



- 1
- 0.5
- 0
- 0.75

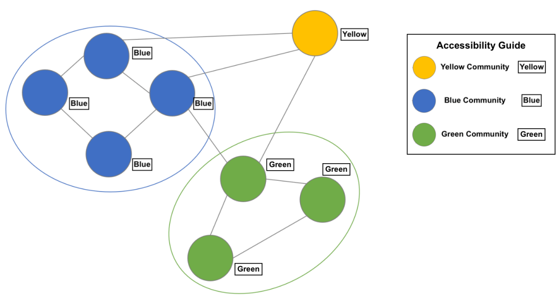
4 Multiple choice 1 point

Which of the following statements about network communities and community detection is/are **TRUE** :

- The clustering coefficient is inversely proportional to the degree of the node for all scale-free networks.
- A signature of nested hierarchical modularity is the inverse relation between a node's clustering coefficient and its degree.
- If we observe an inverse relationship between the clustering coefficient and node degree, we can conclude that the network exhibits hierarchical modularity.
- We can use the dendrogram generated by hierarchical clustering to examine if a network exhibits hierarchical modularity.

5 Multiple choice 1 point

See the following figure. Suppose that SS_{blue} is the similarity between the yellow node and group of blue nodes, measured using single linkage, while SC_{blue} represents the complete linkage, and SA_{blue} represents the average linkage. The similarity between the yellow node and the group of green nodes is denoted, similarly, by SS_{green} , SC_{green} , SA_{green}. Which of the following equation is correct?



- $SS_{blue}= SS_{green}$
- $SC_{blue}=SC_{green}$
- $SA_{blue}=SA_{green}$
- None of above